Supplementary Material

Yield-constrained agrivoltaic potential of India under current and mid-century climate. Sections S1 to S7 give the detail behind Section 2 of the main text, and Section S8 shows the validation of the crop and light models.

S1. Layout geometry and light model

Table S1. Geometry of the agrivoltaic layouts.

Layout	Row axis	Module tilt / azimuth	Module width per row	Lower edge height	Uncultivated strip	Ground strips simulated
South-facing, elevated	East-west	25° / 180°	2.0 m	3.0 m	None	5
East-west ridge, elevated	North-south	10° / 90° and 270°	2 × 2.0 m	3.0 m	None	3, mirrored to 5
Vertical bifacial	North-south	90° / 90° (bifacial)	2.0 m	0.8 m	0.5 m	Pitch mean

Pitch = module width per row / GCR. Simulated GCR: 0.1–0.5 in steps of 0.1 for all crops, plus 0.6 and 0.7 for wheat and pearl millet and 0.05 for chickpea and groundnut.

For a ground point, beam transmission is the fraction of rays at the projected solar zenith angle that pass between the modules of 81 rows centred on the point, and the sky view factor is the visible fraction of an isotropic sky. Both were evaluated at 200 points across the cultivated part of the pitch and averaged over five strips. The east-west ridge is symmetric about the row axis, so three strips were simulated and mirrored. For vertical rows the strip factors were averaged over the pitch, because the morning and afternoon shade sweeps the whole inter-row space. Hourly irradiance used local solar days in Indian Standard Time and the pvlib Linke turbidity climatology.

Table S2. Light model against the analytical two-dimensional functions in pvlib for single-segment rows (26.4°N, 73.1°E; hourly sun positions of 2010 with solar zenith below 85°).

Layout	GCR	Mean sky view factor, this model	Mean sky view factor, pvlib	Largest pointwise difference	Hourly sunlit fraction, MAE
South-facing	0.1	0.9008	0.9009	0.009	0.0013
South-facing	0.2	0.8040	0.8037	0.009	0.0014
South-facing	0.3	0.7089	0.7086	0.012	0.0015
South-facing	0.4	0.6163	0.6162	0.017	0.0014
South-facing	0.5	0.5274	0.5273	0.016	0.0011
Vertical	0.1	0.9047	0.9050	0.005	0.0014
Vertical	0.2	0.8202	0.8198	0.007	0.0012
Vertical	0.3	0.7435	0.7440	0.009	0.0011
Vertical	0.4	0.6768	0.6770	0.008	0.0011
Vertical	0.5	0.6182	0.6180	0.011	0.0009

The east-west ridge has two segments per row and no direct pvlib counterpart; it uses the same ray-tracing code.

S2. Climate data processing

NEX-GDDP-CMIP6 [1] variables were converted as follows: shortwave radiation from W m^{-2} to MJ m^{-2} d^{-1}; precipitation from kg m^{-2} s^{-1} to mm d^{-1}; dew point from mean daily temperature and relative humidity with the Magnus formula, capped at the daily minimum temperature; and wind speed from 10 m to 2 m with the FAO-56 logarithmic profile [2]. Model calendars of 360 or 365 days were mapped onto the real calendar by nearest-day resampling. The climate grid coincides with the simulation grid, so no spatial interpolation was needed.

Table S3. Climate periods and CO_2 concentration.

Period	Years	Scenario	CO_2 (ppm)	Climate models
Reference	1995–2014	Historical	378	GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0, UKESM1-0-LL
Mid-century	2041–2060	SSP2-4.5	507	Same five
Mid-century	2041–2060	SSP5-8.5	563	Same five

Period-mean CO_2 [3], constant within each simulation.

S3. Crop model set-up

DSSAT 4.8 [4,5] was run through the DSSATTools Python interface. Initial soil water was 50% of plant-available water and initial mineral nitrogen was low (0.5 mg NH_4 and 1.0 mg NO_3 per kg soil), and crops were harvested at maturity. Nitrogen was applied as urea incorporated at 5 cm; irrigated wheat received two equal doses at sowing and 30 days after sowing. Rainfed pearl millet and rainfed kharif groundnut were sown automatically in a ±15-day window around the calendar date when soil water in the top 30 cm was 40–100% of plant-available water and soil temperature was 10–40 °C. Soil profiles used the six SoilGrids 2.0 depths [6], sampled at the cell centre; lower limit, drained upper limit, saturation and saturated conductivity came from the Saxton and Rawls functions [7]. The 108 cells (3.6%) without SoilGrids data at the centre took the profile of the nearest valid cell.

Table S4. Management scenarios.

Scenario	Wheat N (kg ha^{-1})	Pearl millet N	Chickpea, groundnut N	Irrigation trigger (share of plant-available water)	Use
Calibrated (main)	30–180 by state	10–30 by state	0	50% (wheat, millet); 20% (legumes)	All periods
Recommended	120 irrigated (60 + 60), 60 rainfed	40	20	50%	Reference period, sensitivity

Triggers apply to irrigated systems, in the top 30 cm of soil.

Table S5. Planting settings.

Crop	Plants m^{-2}	Row spacing (cm)	Sowing depth (cm)	Sowing rule
Wheat	225	22.5	5	GGCMI spring wheat date
Chickpea	33	30	5	Wheat date − 14 d
Pearl millet	11	45	5	GGCMI millet date; rainfed automatic

Crop	Plants m^{-2}	Row spacing (cm)	Sowing depth (cm)	Sowing rule
Groundnut	33	30	5	GGCMI groundnut date; rainfed kharif automatic

Sowing dates from the GGCMI Phase 3 crop calendar [8], irrigated and rainfed calendars separately.

S4. Nitrogen calibration

Simulated open-field cell yields (mean of GFDL-ESM4 and MPI-ESM1-2-HR) were combined into state means weighted by irrigated and rainfed share and by harvested area, and compared with the 1995–2014 mean state yield from the ICRISAT District Level Database [9]. States without enough observed area used the area-weighted median of the calibrated levels. The hill states remained over-estimated at the lowest level; most of their cropland is removed by the slope rule.

Table S6. Calibrated nitrogen level and open-field state yield, reference period.

State	Crop	N (kg ha^{-1})	Observed (kg ha^{-1})	Simulated (kg ha^{-1})
Punjab	Wheat	180	4411	4656
Gujarat	Wheat	180	2553	2854
Haryana	Wheat	90	4172	3739
Rajasthan	Wheat	90	2898	3362
Karnataka	Wheat	90	852	876
Uttar Pradesh	Wheat	60	2761	2652
Madhya Pradesh	Wheat	60	1935	1623
Bihar	Wheat	30	2126	2026
Maharashtra	Wheat	30	1415	1351
West Bengal	Wheat	30	2430	2989
Himachal Pradesh	Wheat	30	1413	3952
Uttarakhand	Wheat	30	1959	3650
Chhattisgarh	Wheat	30	1067	1433
Assam	Wheat	30	1182	1774
Jharkhand	Wheat	30	1421	2281
Other states	Wheat	60	–	–
Gujarat	Pearl millet	30	1355	1129
Haryana	Pearl millet	30	1443	1328
Madhya Pradesh	Pearl millet	30	1406	1173
Uttar Pradesh	Pearl millet	30	1513	1336
Rajasthan	Pearl millet	10	689	637
Maharashtra	Pearl millet	10	758	1183
Karnataka	Pearl millet	10	723	2023
Tamil Nadu	Pearl millet	10	1492	2589
Andhra Pradesh	Pearl millet	10	1277	2059
Other states	Pearl millet	10	–	–

Level 10 kg N ha^{-1} was tested for pearl millet only.

S5. Legume yield-gap factors

Table S7. Observed and simulated open-field legume yields and yield-gap factors (applied to absolute production only).

State	Crop	Observed (kg ha^{-1})	Simulated (kg ha^{-1})	Factor (obs/sim)
Madhya Pradesh	Chickpea	925	854	1.08
Rajasthan	Chickpea	725	522	1.39
Maharashtra	Chickpea	677	436	1.55
Uttar Pradesh	Chickpea	858	1235	0.70
Karnataka	Chickpea	550	1053	0.52
Andhra Pradesh	Chickpea	1096	1292	0.85
Chhattisgarh	Chickpea	746	1089	0.68
Haryana	Chickpea	810	1169	0.69
Gujarat	Chickpea	855	1191	0.72
Telangana	Chickpea	1052	733	1.44
Bihar	Chickpea	1014	820	1.24
Gujarat	Groundnut	1313	2640	0.50
Andhra Pradesh	Groundnut	754	2352	0.32
Karnataka	Groundnut	721	2988	0.24
Tamil Nadu	Groundnut	1979	2654	0.74
Maharashtra	Groundnut	1122	3806	0.30
Rajasthan	Groundnut	1387	2084	0.67
Odisha	Groundnut	1488	3444	0.43
Telangana	Groundnut	1284	1246	1.03
Madhya Pradesh	Groundnut	1176	3665	0.32
Uttar Pradesh	Groundnut	813	3676	0.22

States with at least 50 kha, ordered by area. Area-weighted mean factor: chickpea 1.09, groundnut 0.44.

S6. Aggregation, eligible land and energy

Irrigated and rainfed systems of a crop were combined by production, in the same way as gridded crop-model outputs are aggregated across management systems [8]:

(S1)

where a_w is the share of the crop's area under water regime w and $\bar{Y}_{0,w}$ and RY_w are the open-field yield and RY of that regime. Using the ratio of seasonal means rather than the mean of seasonal ratios in RY keeps seasons with crop failure from dominating. The eligible area of a crop is

(S2)

where A_c is its harvested area (ha) and φ_S the share of the cell with slope ≤ S % (S = 5 in the main case), computed from GLO-90 [10] by finite differences. The rule assumes that cropland is spread evenly over the slope classes of a cell. For single-crop results, capacity is $\eta\, I_{STC}\, A_e\, GCR_{max}$. Annual energy is

(S3)

where Y_{sp} is the specific yield, the final yield of IEC 61724-1 [11] (kWh kW_p^{-1} yr^{-1}), of the cell and layout, interpolated at GCR_{max}. It was computed with pvlib [12] from the same hourly irradiance as the crop simulations, for every fifth year of each GCM (twenty model-years per period). Hourly air temperature followed a cosine between the daily minimum at 06:00 and maximum at 15:00.

Table S8. Parameters of the PV energy model.

Parameter	Value
Module	Bifacial glass-glass, n-type
Efficiency at STC	22%
Bifaciality	0.80
Temperature coefficient of power	−0.30% K^{-1}
Cell temperature	Faiman model [13], pvlib default coefficients, 2 m wind
Irradiance on modules	pvlib infinite sheds, isotropic sky, front and rear
Ground albedo	0.20
DC model	PVWatts [14]
System losses	14%
Inverter efficiency	96%

S7. Implementation and quality control

The workflow was written in Python. Crop simulations, one task per cell, GCM, period and management case, ran on a 12-core cloud virtual machine. Each task wrote its results only on completion, so an interrupted run resumed without duplicates, and DSSAT runs that stopped with an error were retried up to three times. A report every 20 minutes checked three rules and stopped the run on a breach: failed tasks above 2%; median open-field yield of a crop outside a plausible range for India (wheat 800–7000, chickpea 200–3000, groundnut 200–5000 and pearl millet 150–4500 kg ha^{-1}, applied after a full pass over all cells); and more than 5% of irrigated open-field seasons with zero yield.

S8. Validation figures

Figs. S1 to S3 compare the crop model with observed state yields and the light model with an analytical reference and with field measurements. Fig. S1 uses the states with at least 50 kha of the crop. Fig. S2 repeats Table S2 in graphical form. Fig. S3 uses the PPFD measured by Santra et al. [15] at the 105 kW_p agrivoltaic plant of ICAR-CAZRI, Jodhpur, on clear days of 20 to 24 December 2018, 10 cm above the ground. Rows run east-west, modules face south at 26° tilt with the lower edge at 0.5 m, and the three blocks differ in the number of module rows (one, one and a half, two and a half) and in the interspace (2.95, 5.55 and 8.90 m). Measured and modelled values are both relative to the unshaded interspace, which removes the conversion between irradiance and photon flux.

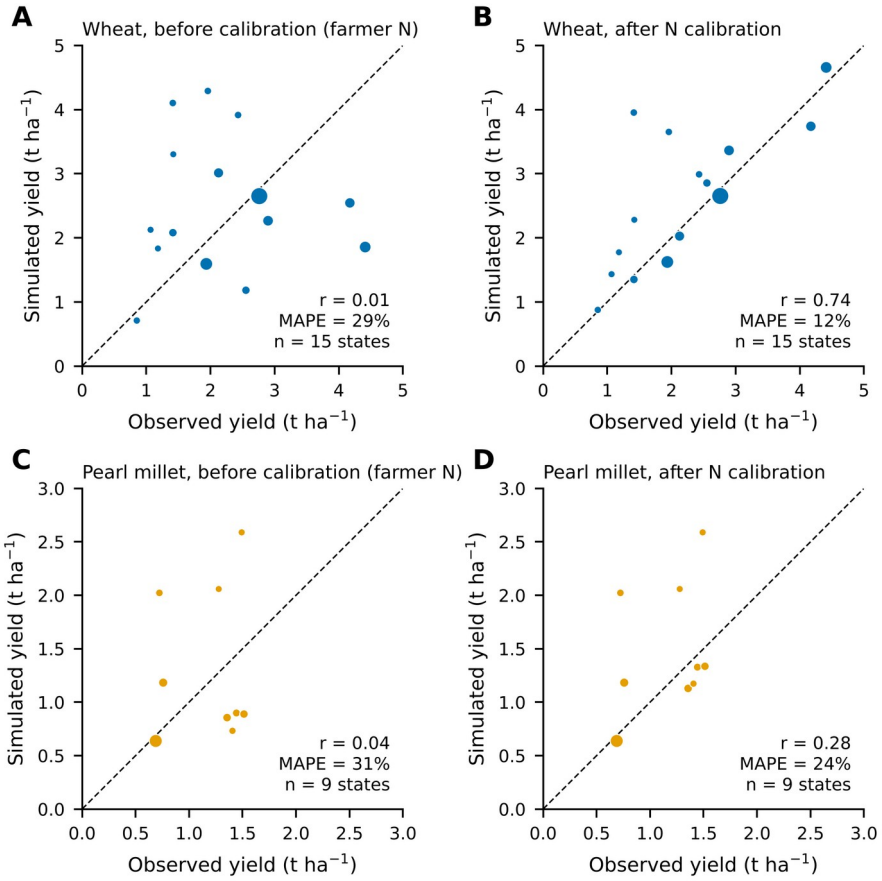


Fig. S1. State mean open-field yields, simulated against observed (ICRISAT District Level Database, 1995–2014), before and after nitrogen calibration. (A, B) Wheat. (C, D) Pearl millet. Before calibration all states used farmer nitrogen (wheat 60, pearl millet 10 kg N ha^{-1}); after calibration each state used its calibrated level (Table S6). Simulated yields are the mean of GFDL-ESM4 and MPI-ESM1-2-HR. Marker area is proportional to observed harvested area; only states with at least 50 kha are shown. The dashed line is 1:1. r is the Pearson correlation and MAPE the mean absolute percentage error weighted by harvested area. Pearl millet in the southern states (Karnataka, Tamil Nadu, Andhra Pradesh) stays over-predicted even at 10 kg N ha^{-1}.

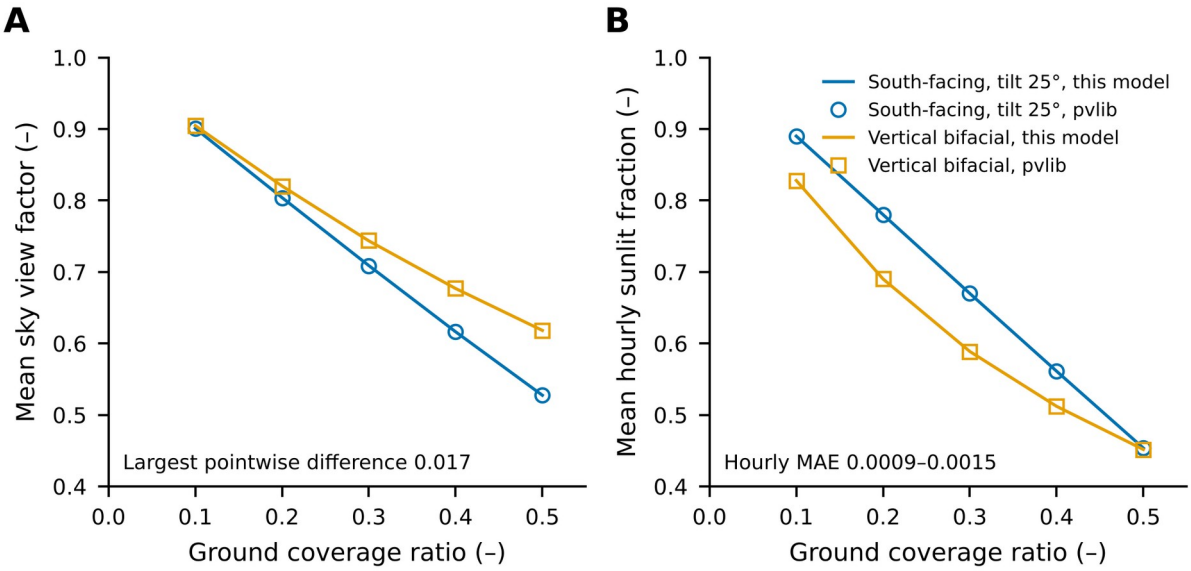


Fig. S2. Light model against the analytical view-factor functions of pvlib for single-segment rows at 26.4°N, 73.1°E (hourly sun positions of 2010, solar zenith below 85°). (A) Sky view factor of the ground averaged across the pitch. (B) Sunlit fraction of the ground averaged over all daylight hours.

Lines are this model and open markers pvlib, for the south-facing layout (tilt 25°) and the vertical bifacial layout. The largest pointwise difference in sky view factor is 0.017 and the mean absolute error of the hourly sunlit fraction is 0.0009 to 0.0015.

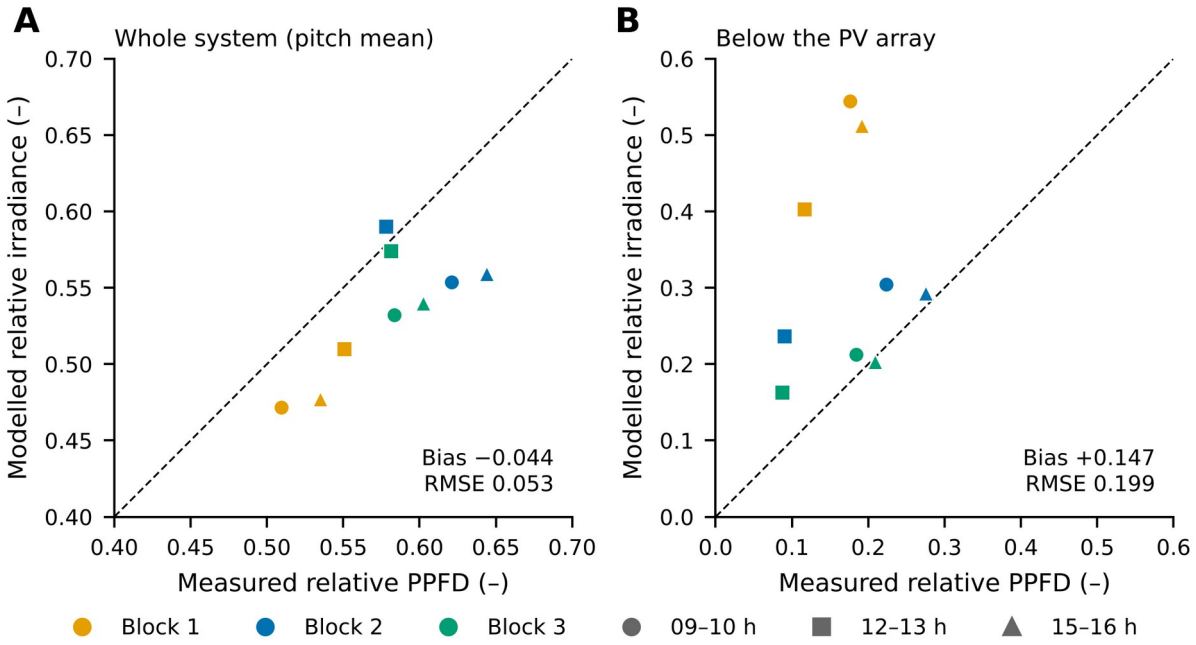


Fig. S3. Modelled relative irradiance against measured relative PPFD at ICAR-CAZRI, Jodhpur [15], by block (colour) and time window (marker). (A) Whole system, averaged over the row pitch: bias −0.044, RMSE 0.053. (B) Ground below the PV array: bias +0.147, RMSE 0.199. The model used clear-sky beam and diffuse irradiance with a broadband diffuse fraction. The dashed line is 1:1. The model reproduces the pitch-mean light but over-estimates the light directly below the modules, most strongly under the single-row Block 1. In late December the noon solar elevation at Jodhpur is about 40°, so below-array light is over-estimated for low winter sun: modelled beam reaches the ground under the low modules more often than measured, and the model includes neither frames, supports nor reflection from the ground and the array. The crop model uses light averaged over strips that span the cultivated pitch, which corresponds to panel A.

Supplementary references

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